Makalakshmi V

Chennai, India
GenAI Developer | AI/ML Engineer | LLM & RAG Specialist| maka23072001@gmail.com | 6383353438 |

Professional Summary

Generative AI Engineer with 3+ years of experience building and deploying production-grade LLM solutions, RAG pipelines, and AI-driven automation systems. Strong expertise in OpenAI, prompt engineering, API integrations, and scalable backend development using Python. Proven ability to design intelligent conversational agents, optimize AI workflows, and deliver measurable business impact.

Skills

AI Automation • Generative AI • LLMs • OpenAI • Azure OpenAI • RAG • Prompt Engineering • LangChain • Embeddings • Vector Databases • Python • FastAPI • REST APIs • MongoDB • Docker • CI/CD • API Integrations • Agentic Workflows • Jira• Webhooks • Cloud Deployment

Professional Experience

AI Automation Specialist – Vendasta Technologies | Chennai| JUN 2025-Present

- Designed and optimized **LLM-powered appointment booking workflows**, improving booking success through advanced prompt refinement and real-time calendar API validation.
- Built and enhanced **GenAI conversational agents** for service qualification, lead capture, and automated booking confirmation across web and SMS channels.
- Applied prompt and context engineering techniques to improve response accuracy and reduce hallucinations in production AI systems.
- Integrated third-party APIs (POS, Calendar systems) for real-time availability checks and seamless appointment scheduling.
- Debugged automation failures (payment links, triggers, lead routing), reducing production errors and improving system reliability.
- Collaborated with cross-functional teams to refine AI workflows and improve automation accuracy and customer conversion rates.
  Tech: OpenAI, LLM APIs, Prompt Engineering, REST APIs, JSON Workflows, POS & Calendar Integrations

AI Automation Engineer – Loissoft Technologies | Remote | 2024

- Developed **LLM-based automation workflows** using n8n and Make.com integrating WhatsApp, Gmail, and CRM platforms.
- Built **RAG pipelines** for contextual lead qualification and document-based AI response generation.
- Automated sales and support processes, reducing manual effort by 70%.
- Integrated FastAPI-based chatbot with real-time speech-to-text using WebSockets.

- Implemented webhook-driven API orchestration for scalable AI system integration.**Tech:** Python, FastAPI, OpenAI, RAG, Embeddings, n8n, Make.com, REST APIs

Software Intern – Highrices Technologies | Chennai | Jan 2024–Dec 2024

- Automated IT workflows using Python and REST APIs.
- Developed internal tools to reduce ticket volume and improve support efficiency.
- Assisted in AI bot deployment, testing, and backend automation.

  Tech: Python, REST APIs

Projects

Enterprise AI Appointment Booking Assistant
Tech: OpenAI (GPT-4), Prompt Engineering, REST APIs, Calendar & POS APIs

Real-Time Speech-to-Text AI Chatbot
Tech: FastAPI, WebSockets, OpenAI, Speech-to-Text APIs

AI Knowledge Assistant (Semantic Search Engine)
Tech: LangChain, FAISS, Python, OpenAI

Education

B.E. Electronics and Communication Engineering – Jeppiaar Institute of Technology – 8.65 CGPA – 2023

Certifications

Prompt Engineering (DeepLearning.AI), Udemy python, Gen Ai,LLM

Languages

English (Fluent), Tamil (Native)